$$r = \frac{17,070 - 16,730}{\sqrt{230 \times 1304}} \qquad r = \frac{340}{547.65} = +0.621$$

Price of the product and supply for the product is positively correlated. When price of the product increases then the supply for the product also increases.

Actual Mean Method:

Ex-1: Estimate the coefficient of correlation with actualmean method for the following data.

Age of Cars in years	3	6	8	9	10	6
Cost of Annual Maintains (in 000 ₹)	1	7	4	6	8	4

Solution:

$$r = \frac{\Sigma xy}{\sqrt{\Sigma x^2 \ \Sigma y^2}} \quad \text{where } x = x-\bar{x},\ y = y-\bar{y}$$

S.No	x	x-$\bar{x}$ x - 7=x	$(x-\bar{x})^2 = x^2$	Y	y-$\bar{y}$ Y −5=y	$(y-\bar{y})^2 = y^2$	xy
1	3	-4	16	1	-4	16	16
2	6	-1	1	7	2	4	-2
3	8	+1	1	4	-1	1	-1
4	9	2	4	6	1	1	2
5	10	3	9	8	3	9	9
6	6	-1	1	4	-1	1	+1
	$\bar{x} \frac{42}{6} = 7$	0	32	$\bar{y} \frac{42}{6} = 7$	0	32	25

$$\Sigma x^2 = 32 \qquad \Sigma y^2 = 32 \qquad \Sigma xy = 25$$

Applying in Formula

$$r = \frac{\Sigma xy}{\sqrt{\Sigma x^2 \ \Sigma y^2}} = \frac{25}{\sqrt{32 \times 32}} = \frac{25}{32} = 0.781$$

r = 0.781, The Car is getting old in years the cost of maintenance is also increasing. The age of Car and its maintenance are positively correlated.

Assumed Mean Deviation Method

Ex 1: Find the Karl Pearson coefficient of Correlation between X and Y from the following data:

X:	10	12	13	16	17	20	25
Y:	19	22	26	27	29	33	37

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